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# STAND-UP

## Standup Meeting



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# REVIEW

## Review the Local UI Code



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# LECTURE

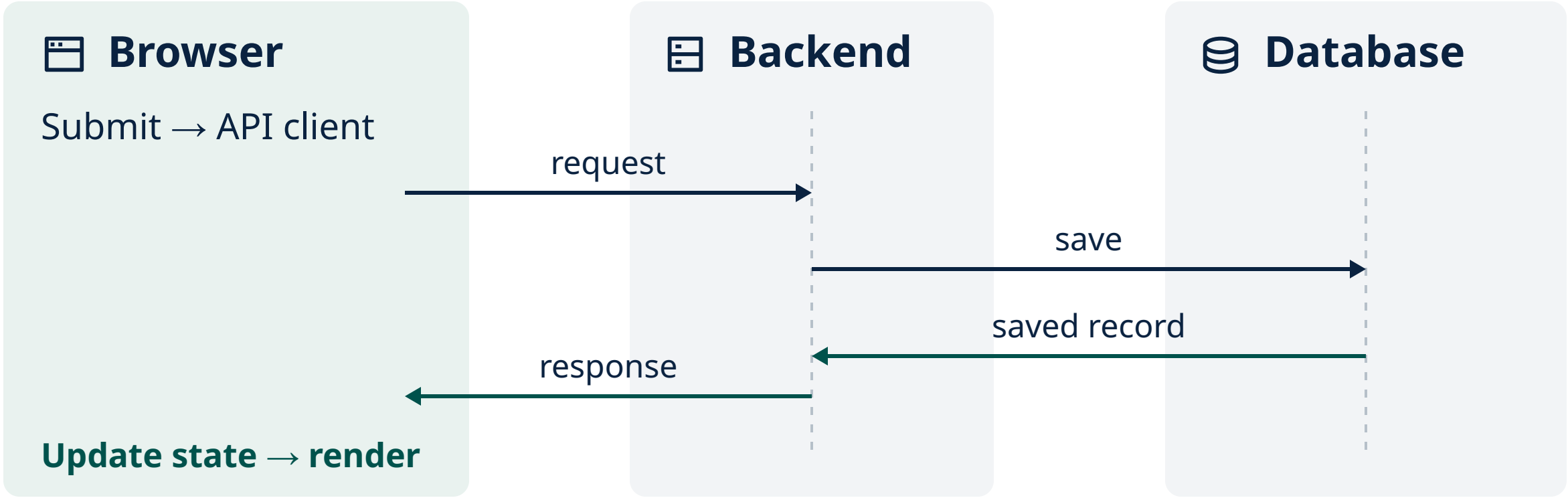
## Explain Asynchronous API Requests



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# Trace one save

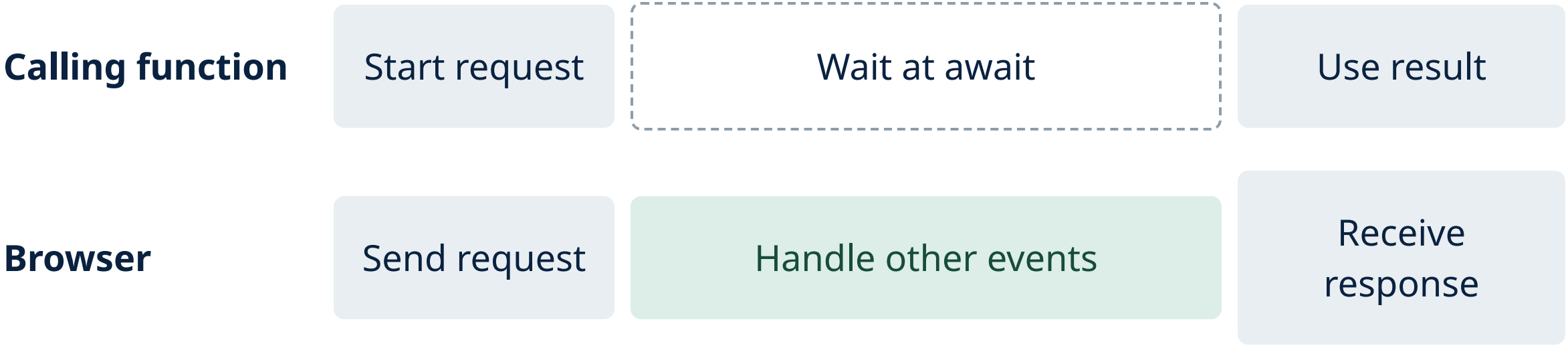
Day 8 state + Day 6 HTTP + Day 7 storage




GET returns an array; POST returns a saved record. Read the returned id and amount_cents.

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# The result arrives later




**Time moves from left to right.**

await pauses the calling function. async does not move synchronous work to another thread.

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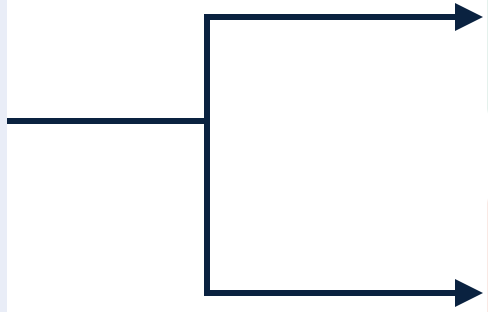
# Wait for the result

Promise states · pending → fulfilled or rejected

`await client.save(…)`

**Pending**

This function waits.



**Fulfilled**

Use the value → update state

**Rejected**

Throw at await → handle the error

Meanwhile, the browser can handle other events.

async functions return a Promise. await pauses this function, not the browser.

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# Response first, JSON next

Teaching excerpt · read the mechanism

```
const response = await fetch('/api/quotes');
if (!response.ok) {
  throw new Error("HTTP " + response.status);
}
const quotes = await response.json();
setItems(quotes);
```

fetch gives a Response. json() gives a Promise for its body. HTTP 4xx needs an explicit status check.

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# Each layer has a job

| Client | Page / form |
| --- | --- |
| Check HTTP status | Use returned data in state |
| Parse success / error bodies | Keep input when saving fails |
| Return values or throw errors | Show feedback and end pending |

**ParcelForm → onAdd → App.add → saveQuote**

Return the Promise through the chain. Which success statements are skipped if await throws?

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# BREAK

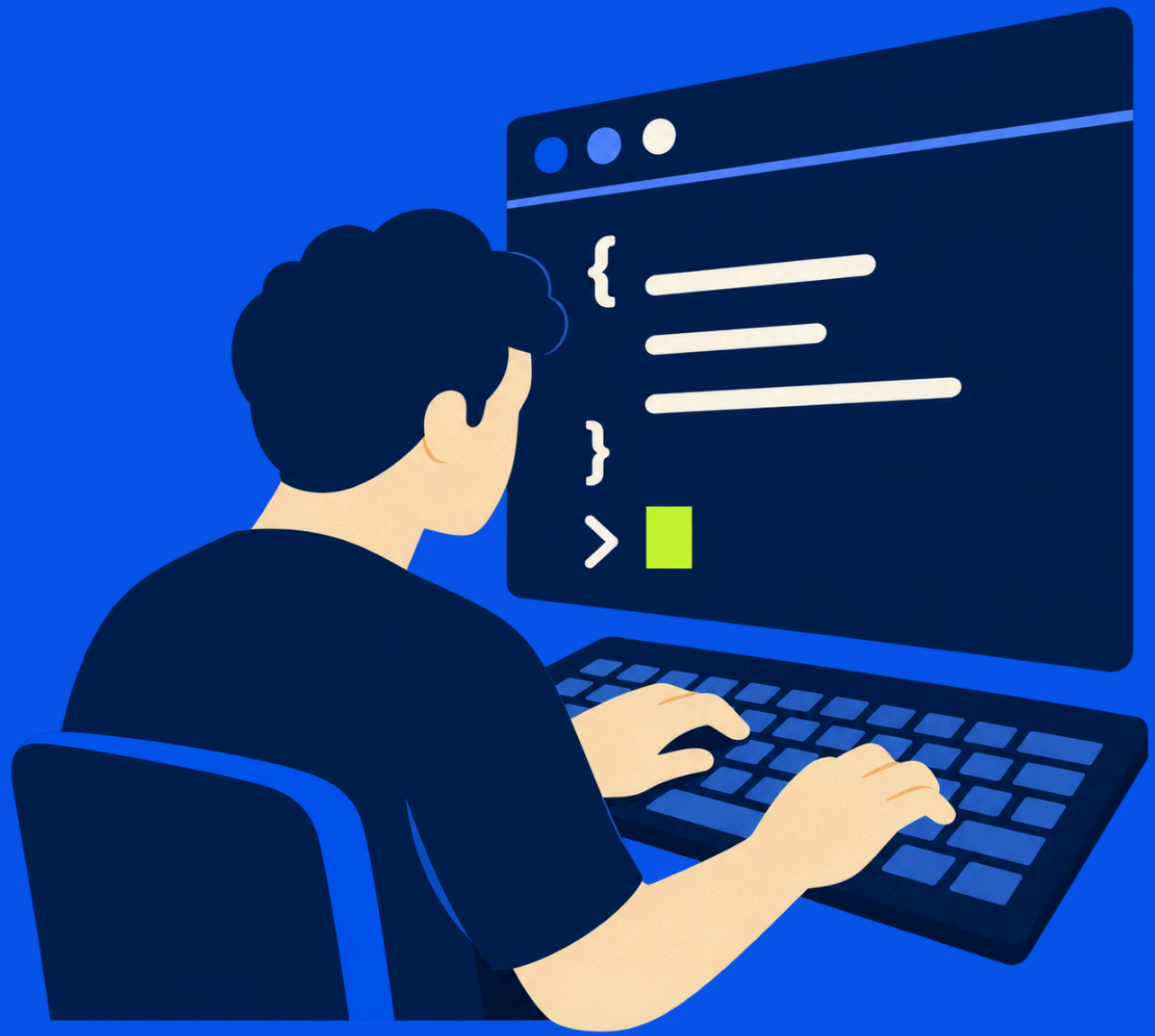
## Take a short break



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# DEMO

## Read a List from the API



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# Read saved records into page state

1. Create the read client in api.ts.
2. Wait for the returned list.
3. Set page state from the response.
4. Compare the response with the displayed records.

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# PRACTICE

## Read One Record by ID Independently

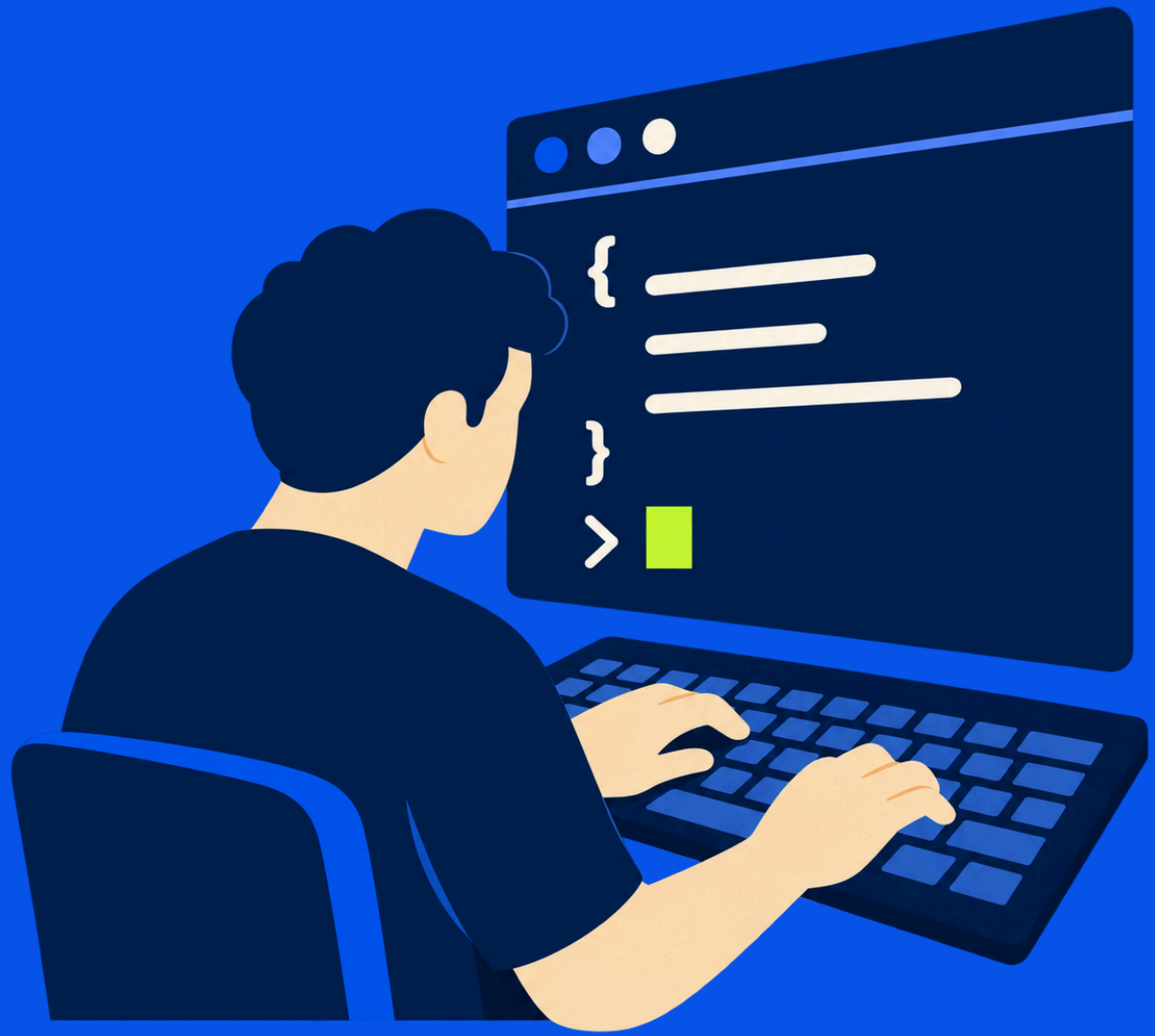


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# DEMO

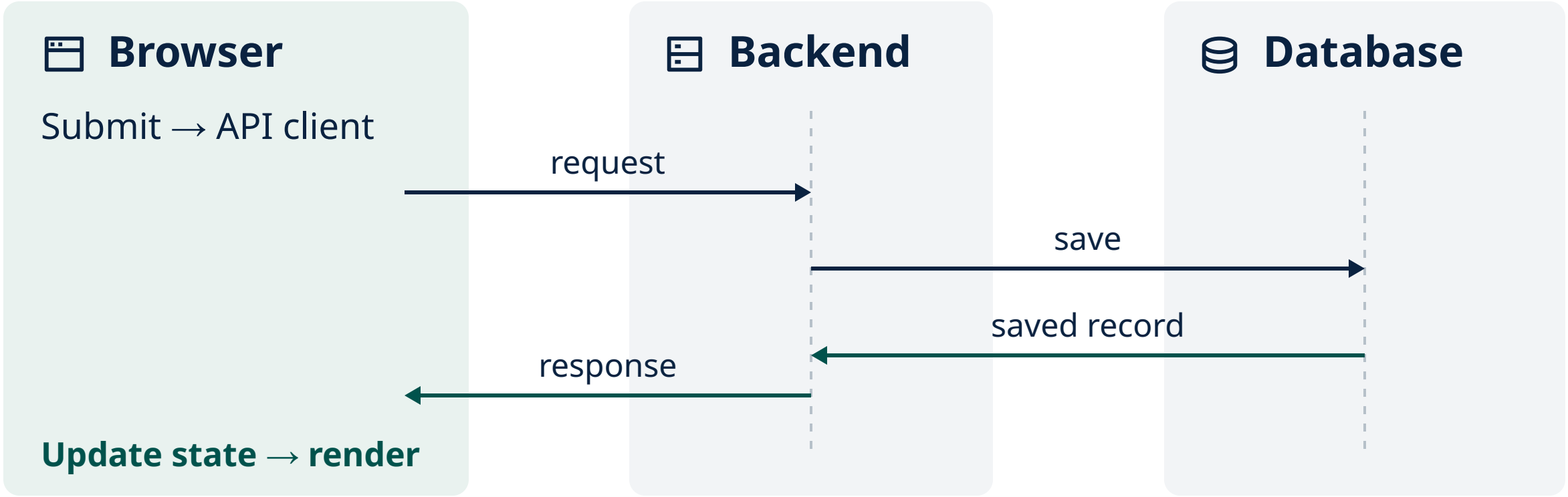
## Connect a Form Submission to the API



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# Trace the submitted fields and returned record

Day 8 state + Day 6 HTTP + Day 7 storage



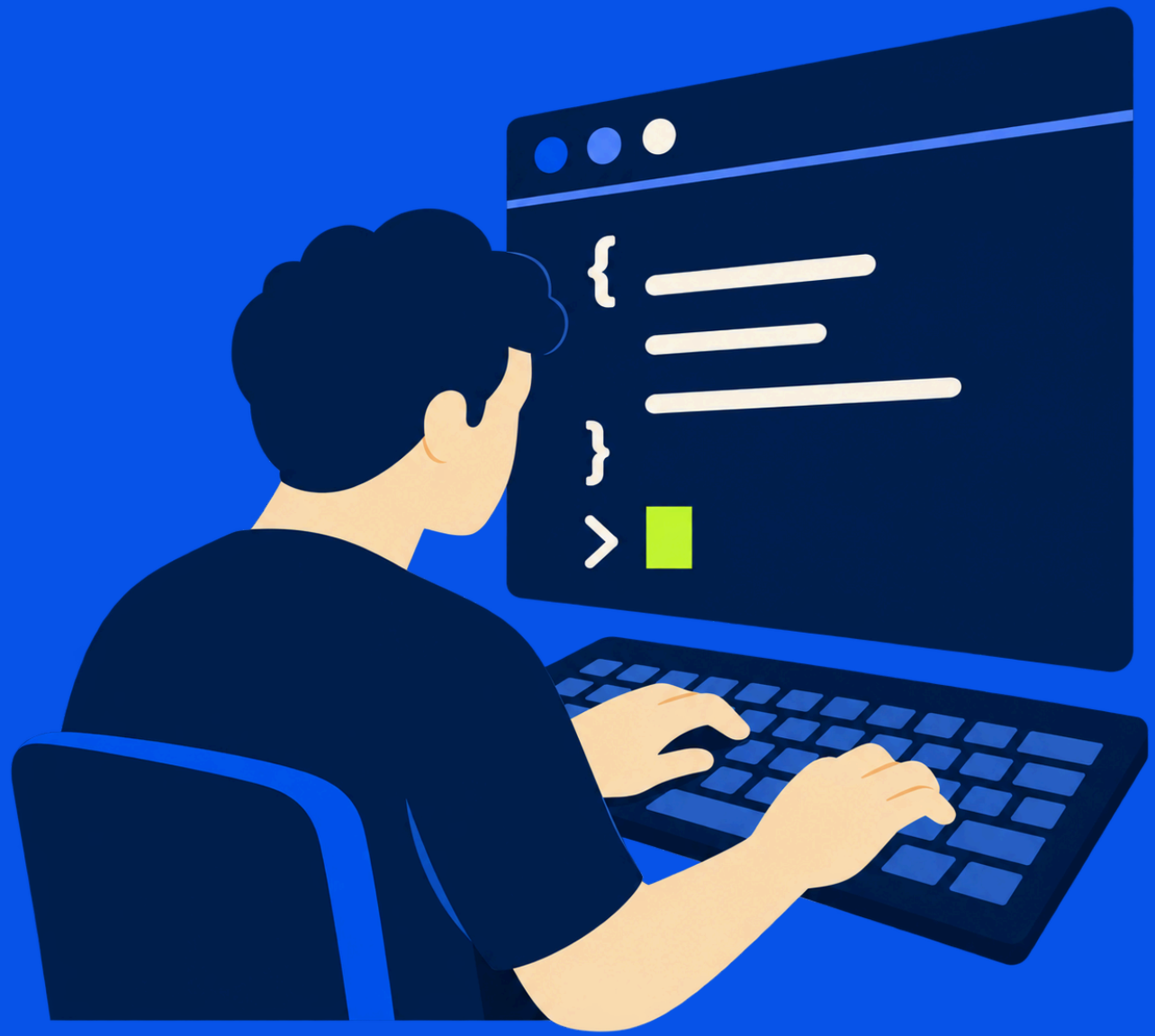

GET returns an array; POST returns a saved record. Read the returned id and amount_cents.

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# DEMO

## Handle Waiting, a Rejected Request (4xx), and Retry



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# Handle waiting, a rejected request and retry

UI sketches · waiting, acceptance and an agreed rejection

### Waiting

Current input

Saving...

Block repeat submit

Previously saved

then

### Accepted

Current input

Save

Show the returned record

Previously saved

Returned record

### Rejected

Current input

Save

Explain the rejection

Previously saved

```
try: await + update        catch: show the error        finally: end loading
```

Check the client’s error contract. Correct → submit → wait again.

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# Check input, then check the rule

Day 8 input checks meet Day 6 server rules

**Browser**

Check input

Help the user correct it.

request →

**Server**

Check input
+ business rules

Accept or reject.

## Browser checks pass ≠ server accepts

Client checks help the user. The server still validates the request.

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# BREAK

## Take a short break



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# PRACTICE

## Check the Completed Request Interaction



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# Check the completed request interaction

UI sketches · waiting, acceptance and an agreed rejection

### Waiting

Current input

Saving...

Block repeat submit

Previously saved

→ then

### Accepted

Current input

Save

Show the returned record

Previously saved

Returned record

### Rejected

Current input

Save

Explain the rejection

Previously saved

```
try: await + update        catch: show the error        finally: end loading
```

Check the client’s error contract. Correct → submit → wait again.

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# LECTURE

## Choose Event or Effect for a Request



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# Choose an event or an Effect

| Trigger | Responsibility |
| --- | --- |
| Click Load saved quotes | Event handler: read again. |
| Mount the page | Effect: synchronize with saved records. |
| Click Save or Delete | Event handler: perform the requested write. |
| Render the component | Describe the UI from current state. |

Keep requests out of the render path.

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# Synchronize when the page appears

Teaching excerpt · read the mechanism

```
useEffect(() => {
  void refresh();
}, []);
```

**Effect**
Start synchronization after commit.

**refresh**
Own the asynchronous read and its feedback.

For this fixed mounting read. The callback is not async. Development may run an extra setup / cleanup cycle.

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# Choose the trigger

Which trigger owns each action? Explain why.

1. The user clicks Reload.
2. The list page first appears.
3. The user clicks Delete.

Then connect the mounting read in the real project.

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# PRACTICE

## Check the Page After Reload



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# Follow the same saved record after reload

Day 7 · Match identity before comparing values

| Save response | = | Database row | = | Later API read | = | Reloaded page |
|---|---|---|---|---|---|---|
| `id` | | `id` | | `id` | | `id` |
| Returned id | | Query that id | | Read that id | | Display that record |

**Reload the page** Keep the backend running.

Same id → compare fields. Reloading the page does not test a backend restart.

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# PRACTICE

## Connect an Existing Delete Action to the API



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# BREAK

## Lunch



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# MINI-GAME

**Optional · A short group game**



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# PRACTICE

## Continue the independent DELETE



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# PRACTICE

## Connect a Service Change to the API Independently



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# BREAK

## Take a short break



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# SHOWCASE

## Compare the UI and API Evidence



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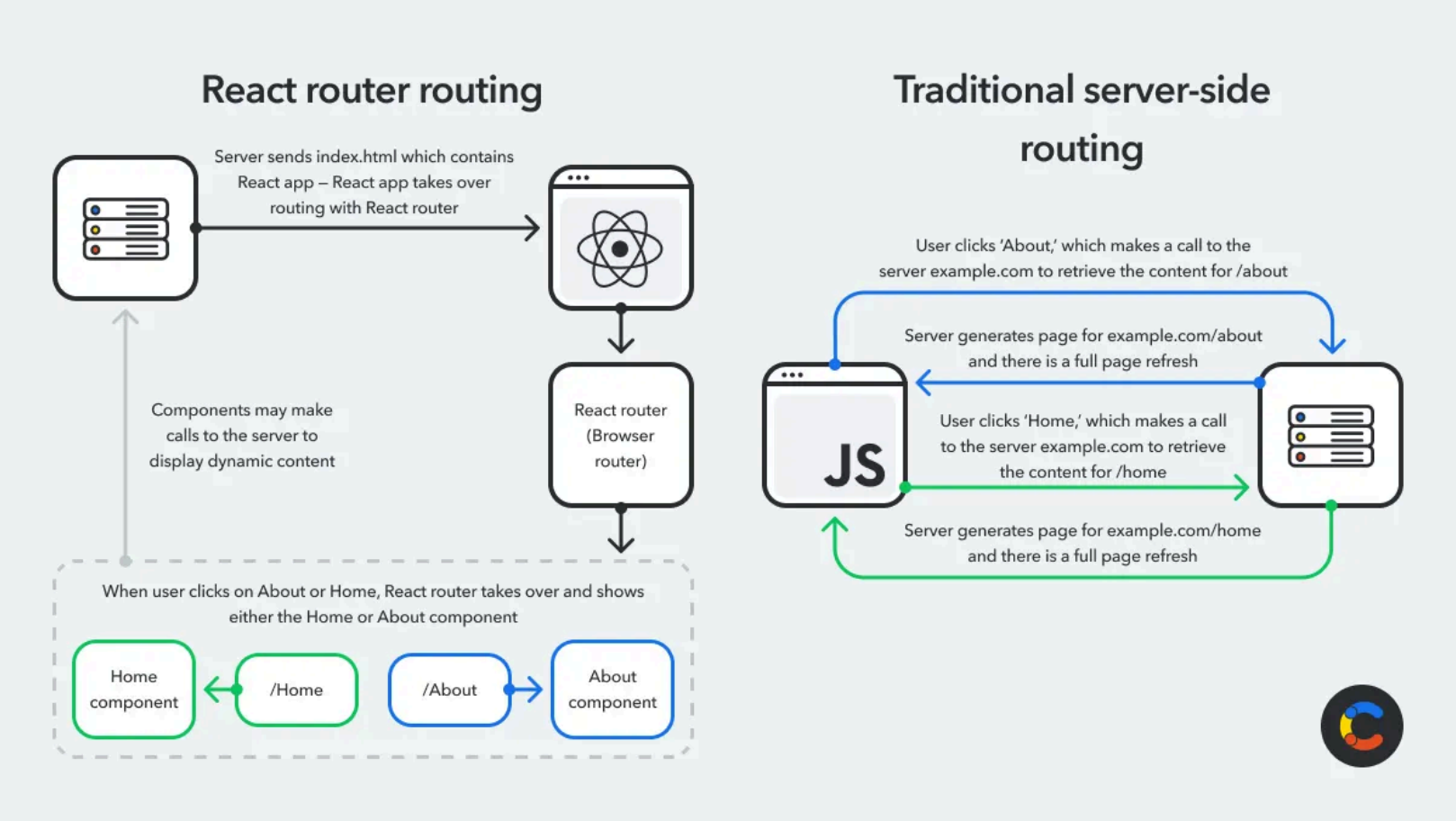
# LECTURE

## Explain URL-Based Pages and Route Parameters



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# Navigation selects a component




2025 Day11, p. 7 · Contentful, React routing · Original illustration

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# A page URL and an API request

### Page URL

`/quotes/17`

BrowserRouter matches a path and selects QuoteDetail.

### API request

`GET /quotes/17`

loadQuote reads the record. The response supplies the data.

Browser requests use the prepared /api proxy. Refresh first requests page resources; the URL alone is not a quote object.

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# Match the URL, then read the data

## 🔀 URL → route match → component

| Path | Page | Responsibility |
| --- | --- | --- |
| `/quotes` | QuoteList | Link to a detail URL |
| `/quotes/:id` | QuoteDetail | Read the string id with useParams |
| `*` | Unknown page | Explain unmatched paths |

BrowserRouter contains Routes and Route. The URL owns id. Server detail and the service draft stay separate.

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# Different failures need different feedback

| What happened? | Who decides? | Feedback |
|---|---|---|
| No matching page path | Frontend router | Unknown page |
| Invalid id format | Detail page | Invalid id; skip API read |
| Valid id, API returns 404 | API, then page | Record not found |
| No response received | Client, then page | Connection failed; retry |

Which case is a route mismatch? Which case needs an API response?

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# DEMO

## Connect List Navigation and Demonstrate Parameter-Based Reading



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# BREAK

## Take a short break



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# PRACTICE

## Build a Detail Page from the URL Independently



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# SHOWCASE

## Compare Navigation, Detail Reads, and Recovery



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# BREAK

## Take a short break

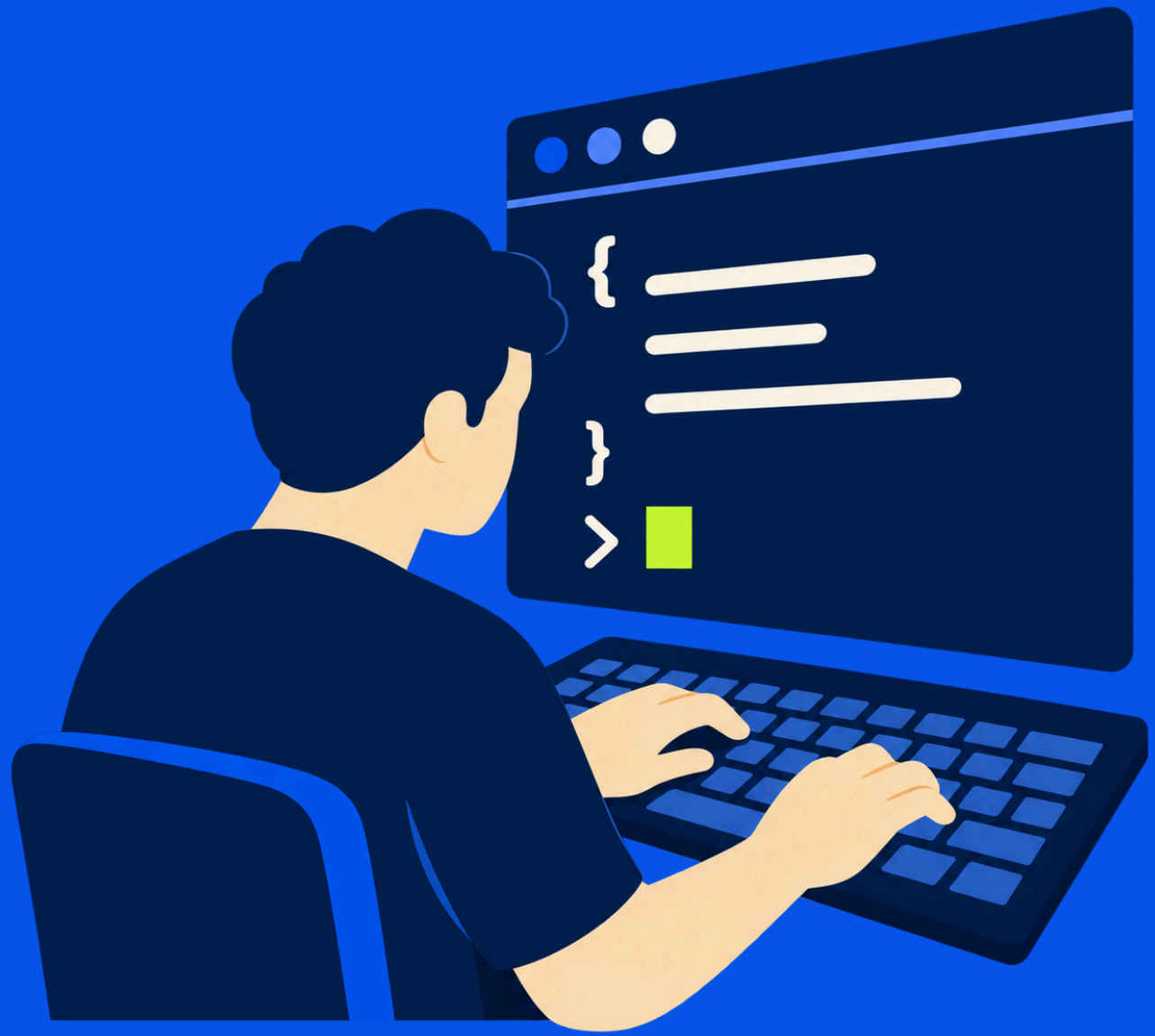


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# DEMO

## Locate a Failure Across System Layers



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# Find where the values diverge

One action · same id, input and version

| | UI | HTTP | Backend | Database |
|---|---|---|---|---|
| **Expected** | ---------- | ---------- | ---------- | ---------- |
| **Observed** | ---------- | ---------- | ---------- | ---------- |

**Where does the first mismatch appear?**

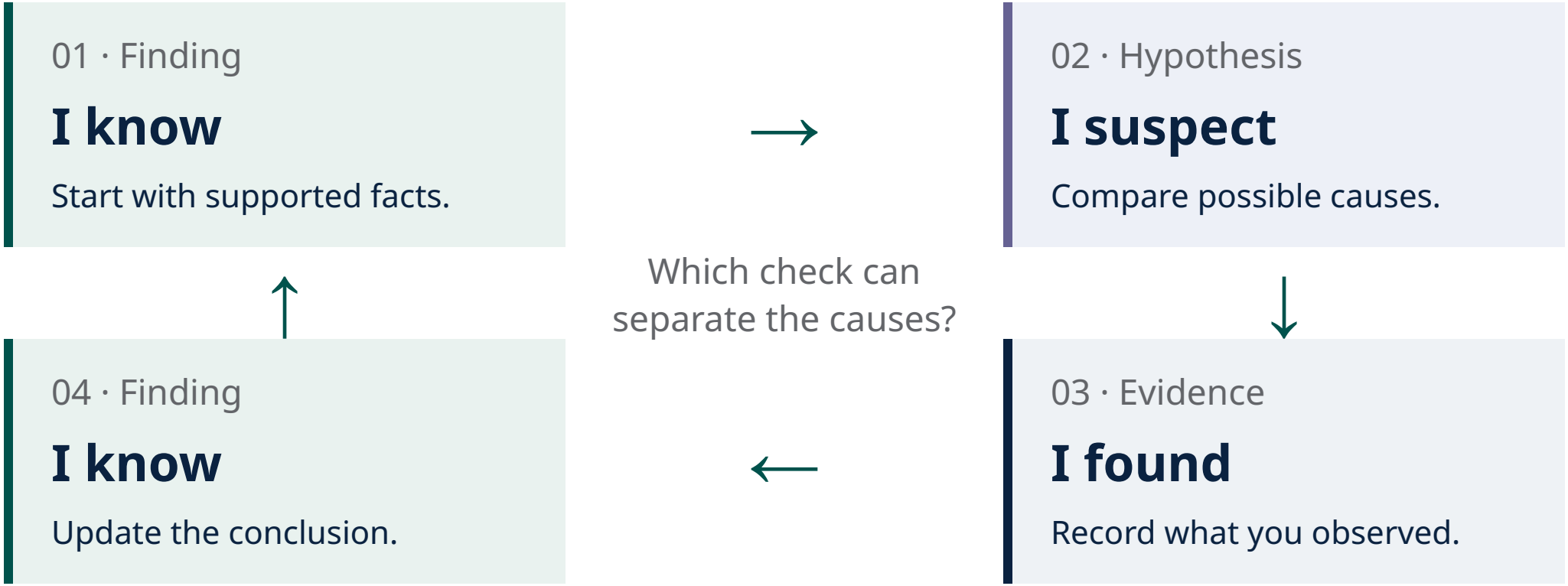
Inspect both sides of that boundary.

Worksheet · fill from the actual trace. A mismatch is a clue, not yet a cause.

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# Use the investigation loop

Day 5 · Same method, more boundaries to inspect




Evidence may support, refute or leave a hypothesis open.

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## Describe What an Existing Test Proves



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# State what the test checks

Read the actual calls and assertions; do not infer scope from a test name.

**Browser UI**

Not exercised

**Illustrative API check**

**Test**

input + assertions

→

**API**

actual calls

→

**Dependencies**

real or substituted?

## Expected

Result + rule source

## Observed

Run result + version

## Limit

Unchecked user path

An API check without a browser does not verify the React update.

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# LECTURE

## Prepare for the Defect Investigation Homework



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